Abstract

As the world transitions to renewable sources of energy, massive grid energy storage will be necessary to accommodate the variable output of these sources. Grid solutions must be efficient, sustainable, affordable, and scalable to keep up with ever-increasing demand. While lithium-ion batteries are the current first choice for grid storage, zinc-ion batteries (ZIBs) are more sustainable and cost-effective. ZIBs have more abundant materials, lower prices, and less potential for harm. This study assesses the performance of zinc-ion batteries made using consumer-grade materials to determine the feasibility of these batteries on a commercial scale. Variations in the composition of the cathode and electrolyte were evaluated for improvements in the discharge profiles and overall performance.

Despite the heavy limitations placed on battery construction by the low-quality materials and simpler design, the batteries were able to effectively charge and discharge without a considerable loss in capacity. It was found that the zinc chloride used in later iterations was a more dangerous and unstable electrolyte, while the addition of manganese dioxide showed substantial improvement to battery performance in Iteration 5. Issues including short-circuiting and gas evolution while recharging demonstrated the need for improved design and construction. Nevertheless, the findings of this experiment underscore the usage zinc-ion batteries for grid storage while emphasizing the need for significant optimizations in the efficiency, stability, and longevity of zinc-ion batteries. Thus, this proof-of-concept study provides a foundation for further research into affordable and sustainable battery technologies.